# MSLM-RF: A Spatial Feature Enhanced Random Forest for On-Board Hyperspectral Image Classification

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***Abstract*— Hyperspectral imaging (HSI) greatly improves the capacity to identify and monitor ground objects due to the high spectral resolution. As the real-time remote sensing monitoring and warning tasks are getting more attention, new algorithms for low-power on-board classification are required to reduce the transmission time of satellite downlink. In this article, we propose the multiscale local maximum random forest (MSLM-RF) to significantly reduce energy consumption while retaining high classification accuracy. The proposed MSLM-RF uses multiscale maximum filters for spatial feature extraction and random forest for classification after spectral and spatial features fusion. The spatial features are efficiently extracted with low computational complexity by regarding the maximum light intensity values in different ranges of pixels as anchor points. MSLM-RF only consists of integer comparisons and a few additions, thereby eliminating the energy-hungry operations such as multiplication and exponentiation. According to experimental results on the HSI benchmark datasets, MSLM-RF delivers a better tradeoff in accuracy and computational complexity than the state-of-the-art classification algorithms. Besides, MSLM-RF gets higher average classification accuracy and lower energy consumption than the previous on-board algorithms. The obtained results show the suitability of the proposed algorithm to accomplish practical real-time classification tasks on-board with low energy consumption.**

***Index Terms*— Computational complexity, hyperspectral image classification, low energy consumption, on-board implementation, random forest (RF), spatial feature extraction.**

## I. Introduction

HYPERSPECTRAL imaging enables a wide range of applications in different fields due to its large amount of information. Fine spectral curves of ground objects can be obtained from the hyperspectral images to accurately distinguish different ground objects. Enabled by the development of image sensors, remote sensing classification tasks for natural disasters or water pollution are getting prevalent and demand real-time processing with high accuracy.

Conventional remote sensing data processing usually takes a relatively long time where the massive volume of raw data collected by sensors is required to be transmitted to the ground station [1] for further processing and distributed to various laboratories. On-board processing is highly desirable for real-time hyperspectral image processing, which eliminates the huge data transfer and unnecessary data processing. Low-level preprocessing methods executed on-board, such as calibration [2], correction [3], compression [4], and georeferencing [5], implement only part of the data preprocessing, while the remaining data processing for tasks is still required to be accomplished on the ground. To realize real-time tasks, high-level algorithms, such as unmixing [6], object detection [7], and classification [8], [9], are required to execute on-board.

For on-board classification tasks, the model is typically trained on the ground and then loaded onto on-board devices for inference. The challenges include efficient on-board implementation of the inference process and high accuracy of the model inference over the new unseen data. In the previous research works for on-board classification, the inferences are implemented on low-power field-programmable gate array (FPGA) [8], [9] and graphics processing unit (GPU) [10]. In [8], extreme learning machines (ELMs) with the input of all bands and input of ten principal components are both evaluated for implementation on FPGA and programmable SoCs. In [9], support vector machine (SVM) was also implemented on FPGA and the classification time meets the AVIRIS real-time requirements. In [10], 2-D convolutional neural networks (CNN2Ds) were implemented on low-power GPU. The runtime and energy cost are compared with running on computer graphics cards. The hardware implementation methods are explored in [8]–[10] to realize a high-performance and real-time hyperspectral on-board classification system. However, the implemented algorithms may still involve computationally expensive operations, such as multiplications and exponentiations, which inevitably lead to high computation energy costs. Due to the limited energy budget of

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satellites, a new hyperspectral on-board classification algorithm is critical for lowering the computational complexity while maintaining high classification accuracy.

The existing hyperspectral classification algorithms can be divided into two categories based on the type of features used for classification. The first category of algorithms [11]–[24] only uses the spectral features of each pixel for classification. These algorithms include probability-based algorithms such as multinomial logistic regression (MLR) [11], [12] and decision tree algorithms [13]–[15]; integrated learning algorithms such as random forest (RF) [16] and gradient boosting decision tree (GBDT) [17]; kernel-based algorithms such as SVM [9], [18], [19]; and artificial neural network algorithms such as multilayer perceptron (MLP) [20], [21], one-dimensional CNN (CNN1D) [22], [23], and ELM [8], [24]. The energy consumption of each algorithm is influenced by the number and type of operations. Most algorithms in [11]–[24] possess a small number of operations and thus generally consume less energy than complex deep neural networks (DNNs). However, the classification accuracies of algorithms in [11]–[24] are typically inferior to the DNN counterpart, which are not sufficient for on-board classification.

In addition to using spectral features, the spatial correlation between pixels in hyperspectral images is considered in the second category of algorithms [25]–[28]. The spatial and spectral features are simultaneously extracted and learned in the one-stage classification algorithms with multidimensional convolutional networks, such as CNN2D [10], [25], three-dimensional CNN (CNN3D) [26], HybridSN [27], and $A^2S^2K$-ResNet [28]. These algorithms in [25]–[28] typically get high classification accuracies, especially for the state-of-art algorithms in [27] and [28]. However, the high accuracies come at the cost of increased computational complexities due to the multiplications involved in massive convolution layers compared to the first category of algorithms in [11]–[24]. Alternatively, there are some works with two-stage spatial–spectral classification algorithms [29]–[34] for better interpretability. In the first stage, the extraction of spatial features is usually completed by using the texture extraction methods such as local binary pattern (LBP) [29]; the multiscale feature extraction methods such as extended morphological profiles (EMPs) [30], extended multivariate attribute profile (EMAP) [31], and Gaussian pyramid (GP) [32]; or the backbone part of deep learning networks such as DeepLab [33] and DCN [34]. Dimensionality reduction methods, such as principal component analysis (PCA) [35] or linear discriminant analysis (LDA) [36], are typically employed in previous spatial feature extraction methods. Then, classifiers, such as SVM or RF, are typically used in the second stage for classification. The two-stage classification algorithms provide high flexibility to separately optimize the feature extraction methods and the classifiers. The two-stage algorithms are therefore considered to be suitable for on-board classification tasks. However, only part of the spatial features can be extracted by the methods in [29]–[31]. The spatial feature extraction method can be further improved, and the selected classifier can be evaluated to achieve a better tradeoff between low computational cost and high accuracy.

In this article, we propose a multiscale local maximum random forest (MSLM-RF) algorithm with two-stage targeting for the on-board hyperspectral image classification with high accuracies and low energy consumption. The multiscale maximum filters are employed in the first stage of MSLM-RF for spatial feature extraction, while RF with a small number of operations is selected in the second stage for classification. The contributions are summarized as follows.

1) A new spatial feature extraction method named multiscale local maximum (MSLM) is proposed based on the properties of hyperspectral images and principles of spatial feature representation. Multiscale maximum filtering effectively integrates local and global spatial features by extracting spatial anchors over different ranges.
2) The optimization space for reducing the computational complexity and energy consumption of on-board classification is fully explored from the algorithm perspective in this article. The basic operation types of proposed MSLM-RF only consist of 8-bit integer comparisons and additions where the power-hungry multiplications and exponentiations are eliminated.
3) By simultaneously considering the classification performance and energy consumption, MSLM-RF provides a better tradeoff between accuracy and computational cost compared with the state-of-the-art algorithms. MSLM-RF also gets higher classification accuracies and lower energy consumption than previous on-board algorithms.
4) Two extension algorithms of the proposed MSLM-RF are explored to further reduce the computational complexity while maintaining competitive classification accuracies.

The remainder of this article is organized as follows. Section II describes the whole framework and basic methods of the proposed MSLM-RF. The computational complexities of different algorithms are also compared and analyzed. In Section III, the classification accuracy, the number of operations, and energy consumption of different algorithms are comprehensively evaluated. Two extension algorithms of MSLM-RF are also presented in Section III. Finally, the conclusions are provided in Section IV.

## II. Proposed MSLM-RF Algorithm

The whole framework and basic methods of the proposed two-stage classification algorithm, named MSLM-RF, are described in this section. The two-stage classification framework of MSLM-RF is described in Section II-A. The spatial feature extraction method based on MSLM is proposed for the first stage and described in Section II-B. The RF for classification in the second stage is reviewed in Section II-C, where the classification accuracy is significantly improved by combining both spatial and spectral features in the proposed MSLM-RF. The computational complexities of previous hyperspectral classification algorithms and the proposed MSLM-RF are analyzed in Section II-D.

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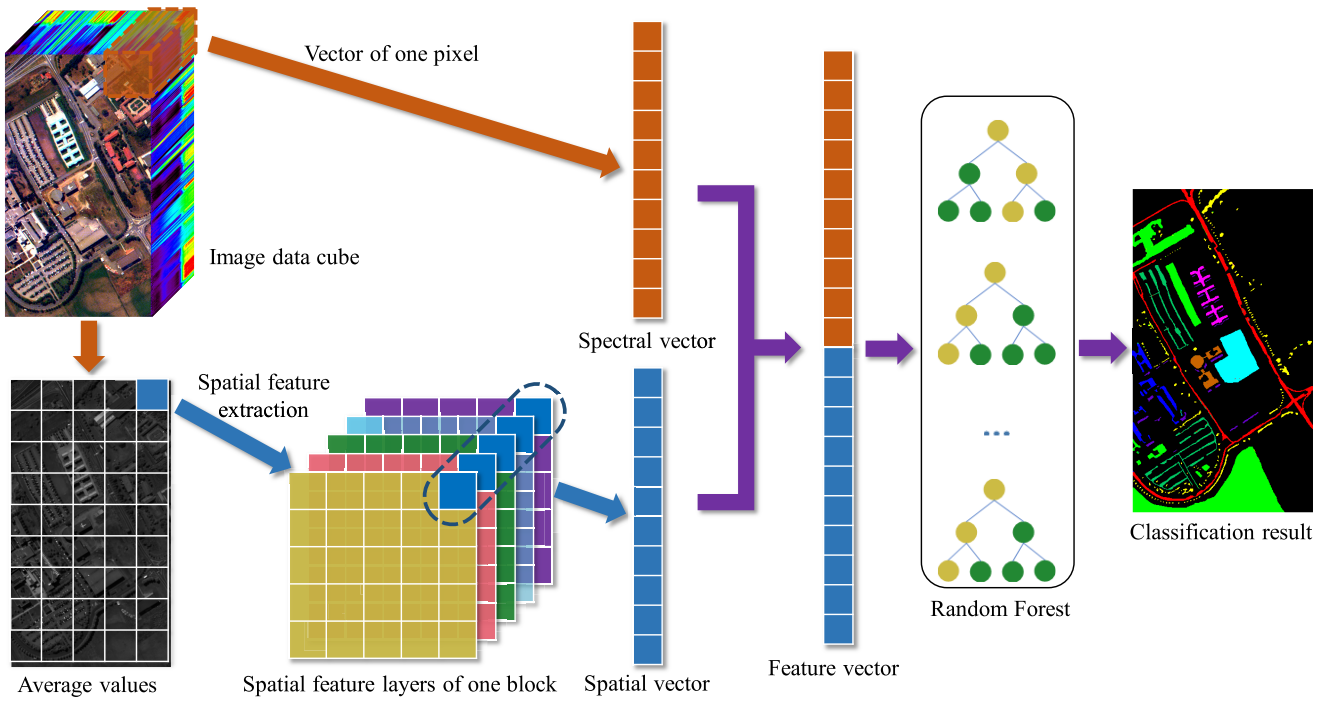



Fig. 1. Framework of MSLM-RF for on-board hyperspectral image classification.

### A. Proposed Classification Algorithm Framework

The whole framework of MSLM-RF is shown in Fig. 1. The satellite collects hyperspectral images of large sizes, and image blocks of an appropriate size, such as 128 × 128 or 256 × 256, are produced by slicing. Then, the spatial feature extraction method is applied to each image block. The spatial feature vector from the output layers and the spectral vector from the hyperspectral image are spliced to obtain the feature vector of the pixel. We use the feature vectors of selected samples to train the RF model. The main computation types in both MSLM and RF are data comparisons that consume less energy than multiplications and exponentiations. Besides, data comparisons have better tolerance for noises or fluctuations caused by cosmic rays than multiplication in DNNs when the algorithm is deployed on-board.

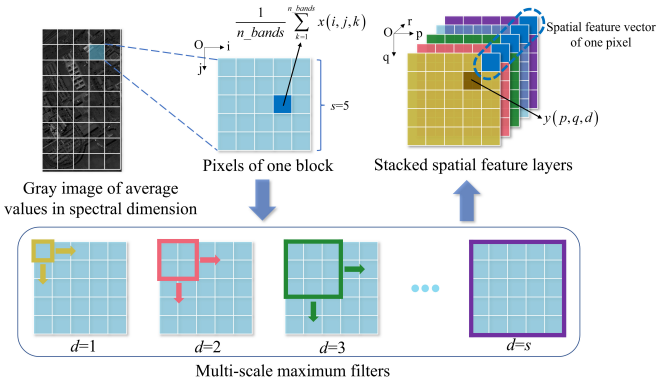



Fig. 2. Proposed spatial feature extraction method with multiscale maximum filters. $d$ represents the size of filter windows. The actual satellite imagery is usually very large and needs to be divided into blocks as the basic processing image unit. An image block with size $s = 5$ is taken as an example.

### B. Proposed Spatial Feature Extraction Method

In Fig. 2, a spatial feature extraction method based on MSLM is proposed in the first stage. MSLM uses maximum filtering to extract spatial anchor points in different ranges of reference pixels, which can be regarded as the main components of image texture features. By combining feature values of different scales into spatial feature vectors, the local and global spatial features of the image are effectively integrated. The entire calculation process of MSLM only consists of comparison operations, which significantly reduces the energy consumption compared to previous feature extraction methods with multiplication operations.

There are hundreds of continuous and subdivided spectral bands in the hyperspectral image and dimensionality reduction is required prior to the spatial feature extraction. Due to the singular value decomposition in PCA or LDA, a large computational cost will be generated in the iterative process. The average values of pixels in spectral dimension are first calculated as the dimensionality reduction instead in MSLM. The average values pass maximum filters with different scales to obtain several layers of spatial features.

Maximum filters are used as the basic operator for feature extraction because the local extremum or "peak and valley" can represent the spatial features of the image as the expression of the image texture [37]. As the image noise in remote sensing is more significant than natural photographs, using maximum values to represent the texture features is less likely to be affected than using the minimum values. The fine texture has many local maxima separated by small distances, while the rough texture has a small number of local maxima separated

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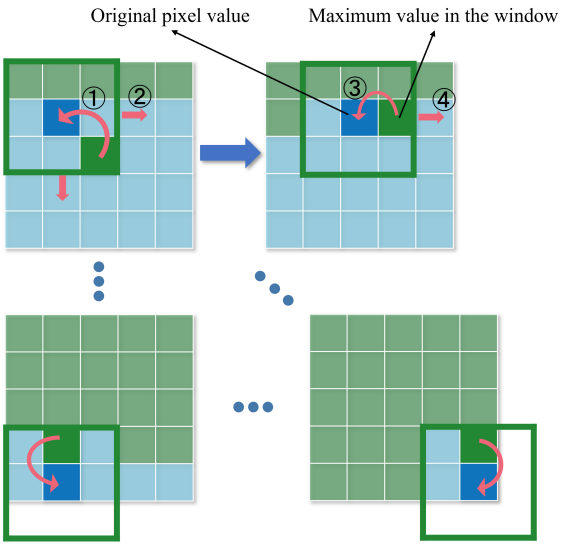



Fig. 3. Pixel replacing process in each maximum filter. $d$ is assumed to be 3.

by large distances. The process of each maximum filter is shown in Fig. 3. The pixel value at the center is replaced by the maximum value in each filter window. The window then slides right and down to construct an entire spatial feature layer. In the actual classification tasks, the maximum filters at the edge of the image should be calculated on pixel values of adjacent blocks.

Next, multiscale methods are typically employed to enhance the feature extraction capability. The traditional methods, such as LBP [29], multiscale CNN [38], EMP [30], and EMAP [31], only extract the spatial features within the filtering or convolutional windows with limited sizes. Specifically, LBP performs comparisons in $3 \times 3$ or $5 \times 5$ neighborhoods. The sizes of kernels used in the multiscale CNN are also restricted to $3 \times 3$ and $5 \times 5$. Several morphological properties are obtained in EMP and EMAP with a limited number of opening and closing operations. In contrast, the pyramid structure is utilized in this work to simultaneously extract local and global spatial features. Inspired by the spatial pyramid matching which takes "spatial pyramid" as an orderless bag-of-features image representation [39], the maximum filters of different scales ($d$ is increased from 1 to $s$) are used to construct spatial pyramid for each pixel, as shown in Fig. 2. Fig. 4 shows the structure of applying multiple filters with the same center position to the targeted pixel. The coordinate of the center point is taken as $(d-1)/2$ for odd $d$ and $d/2$ for even $d$. Multiple spatial feature layers are generated by the proposed multiscale filters, thereby improving the accuracy of the whole two-stage classification algorithm.

Finally, the maximum values in different ranges are used to establish the spatial feature vector of the center pixel since the pixels of remote sensing images are spatially correlated. The establishment process is based on the context classification principle that adjacent pixels are more likely to belong to the same class. The spatial feature of each pixel can be represented by a vector of the pixel values in the specified neighborhood. Four- or eight-connected $3 \times 3$ neighborhoods are generally

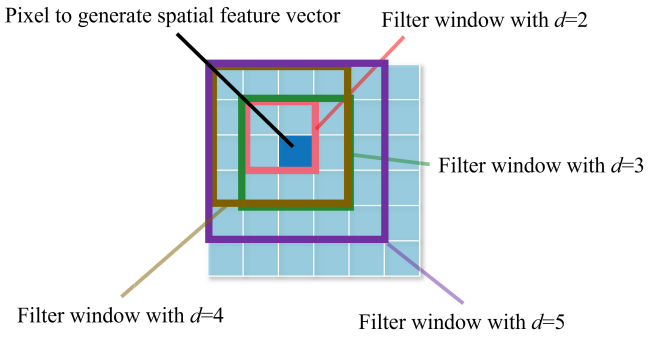



Fig. 4. Applying maximum filters with different window sizes on the same center pixel.

used in the conventional computer vision field [40]. In the proposed spatial feature extraction method shown in Fig. 2, the largest filter window covers the entire image block, not limited to the $3 \times 3$ neighborhood. The output layers of all maximum filters are stacked to create spatial feature vectors instead of using all the values in a fixed neighborhood. The values of pixels farther from the center point are less likely to change the filtering result for the same target pixel, which conforms to the spatial correlation of image pixels. The maximum values can be regarded as anchor points to locate the current pixel in space.

The whole process of the proposed spatial feature extraction method can be summarized as calculating the $d$th value of the output spatial feature vector in position $(p, q)$ as follows:

$$y(p,q,d) = \max_{p-m\le i\le p+n, q-m\le j\le q+n} \left( \frac{1}{n_\text{bands}} \sum_{k=1}^{n_\text{bands}} x(i,j,k) \right)$$
$$m = \left\lceil \frac{d-1}{2} \right\rceil, \quad n = \left\lfloor \frac{d-1}{2} \right\rfloor, \quad 1 \le d \le s \tag{1}$$

where $x(i, j, k)$ is the $k$th value of the pixel spectral vector in position $(i, j)$, and $n_$bands is the number of spectral bands. The filter windows have sizes ranging from 1 to the size (defined as $s$) of the image block and slide in two dimensions. Each pixel is assigned a spatial feature vector containing the maximum values in different scales. The feature vector formed by fusing the spatial and spectral vectors will be used in training RF model and classification, as shown in Fig. 1.

### C. Random Forest

RF is one of the most frequently used classification algorithms in remote sensing [41]. RF is an ensemble learning method made up of multiple decision trees, as shown in Fig. 5. Each decision tree is a weak classifier to classify randomly selected features. The RF integrates all results from trees and the category of the major vote or calculated by averaging is designated as the final output. RF corrects the decision tree's defect of overfitting to training samples and usually gets higher accuracy.

Compared with other classifiers in [11]–[24], RF consumes less energy due to a smaller number of operations in the inference stage, which will be discussed in Sections II-D and III. In particular, RF primarily consists of comparisons, avoiding

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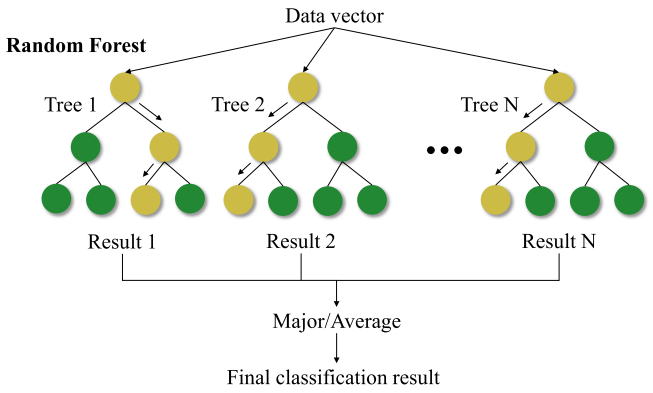



Fig. 5. Structure of RF.

the energy-hungry multiplications in the fully connected layers and exponentiations in Softmax of CNN counterpart. Besides, the classification accuracy of RF can be improved by the proposed MSLM method. After the fusion of spatial and spectral features, RF can get classification accuracy close to those of DNNs. Therefore, RF is chosen as the classifier of MSLM-RF.

### *D. Computational Complexity Analysis*

The statistical formulas calculating the number of operations for each pixel in the classification process of different algorithms are listed in Table I. The operation types include floating-point exponentiation (Exp), floating-point multiplication (Mul), floating-point addition (Add), and integer comparison (Cmp). To prove the effectiveness of the proposed MSLM-RF, the computational complexity of different representative hyperspectral classification algorithms from three categories is included for comparison. For spectral algorithms, we selected the commonly used MLR [11], SVM [18], RF [16], GBDT [17], MLP [20], CNN1D [22], and ELM [24]. For one-stage spatial–spectral algorithms, CNN2D [25], CNN3D [26], and some state-of-the-art algorithms, including HybridSN [27] and $A^2S^2K$-ResNet [28], are chosen. Since the one-stage spatial–spectral classification algorithms are mainly composed of convolutional layers, the number of operations for each pixel passing through a single convolutional layer is shown in Table I. The total numbers of operations for the algorithms can be calculated by accumulating the numbers of multiple convolutional layers. For two-stage spatial–spectral algorithms, we picked EMAP-RF [31] with the same classifier as MSLM-RF. The LBP-RF, EMP-RF, and SPCA-GPs-RF are also constructed by replacing the original classifiers in LBP-ELM [29], EMP-MLP [30], and SPCA-GPs-SVM [32] with RF, respectively, for a fair comparison to the proposed MSLM-RF. In EMP-RF, EMAP-RF, and SPCA-GPs-RF, the dimensionality reduction methods PCA, LDA, and segmented PCA (SPCA) are employed, which account for significant portions of the total number of operations. Since the numbers of operations in dimensionality reduction methods depend on software implementation and calculation iterations for specific inputs, the numbers of multiplications and additions consumed by the dimensionality reduction methods are represented as the functions Mul(PCA/LDA/SPCA) and Add(PCA/LDA/SPCA), respectively, as listed in Table I.

TABLE I
CALCULATION FORMULAS FOR THE NUMBER OF OPERATIONS

| | |
|---|---|
| **Spectral algorithms** ($k$: number of features, $l$: number of classes) | |
| MLR | $kl$ Muls, $kl$ Adds |
| SVM | $M$ Exps, $(kM + 2M)$ Muls, $(kM + l)$ Adds ($M$: number of support vectors) |
| RF | $N$ Adds, $DN$ Cmps ($D$: average depth of trees, $N$: number of trees) |
| GBDT | $N$ Adds, $iDN$ Cmps ($i$: number of iterations, $D/N$: same as RF) |
| MLP | $(kn + nl)$ Muls, $(kn + nl)$ Adds ($n$: number of neurons in hidden layer) |
| CNN1D | $(k - q + 1)ql$ Muls, $(k - q + 1)ql$ Adds ($q$: number of values in each filter) |
| ELM | $(km + ml)$ Muls, $(km + ml)$ Adds ($m$: number of selected neurons in hidden layer) |
| **One-stage spatial-spectral algorithms** | |
| Each convolution layer of CNN2D, CNN3D, HybridSN [27], and $A^2S^2K$-ResNet [28] | $K^2 \times C_{in} \times C_{out}$ Muls, $K^2 \times C_{in} \times C_{out}$ Adds ($K$: kernel size, $C_{in}$ : number of input channels, $C_{out}$ : number of output channels) |
| **Two-stage spatial-spectral algorithms** ($k$: number of features) | |
| LBP-RF | $N$ Adds, $(kp + DN)$ Cmps ($p$: number of points in LBP, $D/N$: same as RF) |
| EMP-RF | Mul(PCA) Muls, (Add(PCA)+$N$) Adds, $(\sum_{1}^{m} S_i^2 + DN)$ Cmps (Mul(PCA)/Add(PCA): number of Mul/Add in PCA, $S_i$ : size of each structuring element, $m$: number of structuring elements, $D/N$: same as RF) |
| EMAP-RF [31] | (Mul(PCA)+Mul(EMAP)+Mul(LDA)) Muls, (Add(PCA)+Add(EMAP)+Add(LDA)+$N$) Adds, $DN$ Cmps (Mul(PCA)/Add(PCA): number of Mul/Add in PCA, Mul(EMAP)/Add(EMAP): number of Mul/Add in calculating attributes, Mul(LDA)/Add(LDA): number of Mul/Add in LDA, $D/N$: same as RF) |
| SPCA-GPs-RF | 2Mul(SPCA) Muls, (2Add(SPCA)+$N$) Adds, $(kK^2 + DN)$ Cmps (Mul(SPCA)/Add(SPCA): number of Mul/Add in SPCA, $K$: kernel size of Gaussian filter, $D/N$: same as RF) |
| **MSLM-RF** | $N$ Adds, $(d_{max}^2 + DN)$ Cmps ($d_{max}$ : size of largest filter, $D/N$: same as RF) |

For spectral algorithms, as each tree randomly selects part of the features as input data, the average depth $D$ is usually smaller than the number of input features $k$ in RF and GBDT.

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TABLE II
PARAMETERS OF THREE DATASETS

| Name | Indian Pines | Pavia University | Salinas |
|---|---|---|---|
| Size | 145×145 | 610×340 | 512×217 |
| Bands | 200 | 103 | 204 |
| Classes | 16 | 9 | 16 |
| Labeled pixels | 10249 | 42776 | 56975 |

Besides, the number ($N$) of trees is typically tens or hundreds in RF and GBDT. Nevertheless, GBDT needs extra number of comparisons for iterations compared to RF. Using RF as the classifier can therefore get a lower computational complexity with smaller number of operations than other spectral algorithms, as listed in Table I.

For spatial–spectral algorithms, the total numbers of operations in CNNs are usually high due to a large number of accumulated operations in multiple convolution layers. Alternatively, the numbers of operations in LBP-RF and the proposed MSLM-RF are smaller than other evaluated spatial–spectral algorithms listed in Table I. In summary, RF, GBDT, LBP-RF, and the proposed MSLM-RF get lower computational complexities with comparison operations compared to other algorithms containing exponentiations or multiplications.

## III. EXPERIMENTS AND RESULTS

In this section, the performance of MSLM-RF on five hyperspectral datasets is comprehensively evaluated, including Indian Pines, Pavia University, Salinas, HyRANK, and Houston. The datasets are briefly introduced in Section III-A. The classification accuracies of different algorithms and the proposed MSLM-RF are compared in Section III-B. The cross-scene classification performance of MSLM-RF is also studied in Section III-C for real-world on-board classification tasks. The effectiveness of the proposed spatial feature extraction method under different conditions is verified in Section III-D. The numbers of operations and energy consumption of different algorithms in the classification process are shown in Sections III-E and III-F, respectively. Finally, we proposed two extension algorithms and evaluated their tradeoffs between accuracy and computational complexity in Section III-G.

### A. Hyperspectral Datasets

Three datasets widely used in hyperspectral image classification are chosen to compare the classification performance of different algorithms: the Indian Pines, Pavia University, and Salinas. The Indian Pines and Salinas datasets were gathered by the AVIRIS sensor in Northwestern Indiana, while the Pavia University dataset was acquired by the ROSIS sensor in Pavia, Northern Italy. The parameters of three datasets are listed in Table II and the ground truths are shown in Fig. 6.

Another two publicly available hyperspectral datasets are selected to test the cross-scene classification performance of the proposed MSLM-RF: the HyRANK and Houston.




Fig. 6. Ground truth and legend of each hyperspectral dataset. (a) Indian Pines. (b) Pavia University. (c) Salinas.

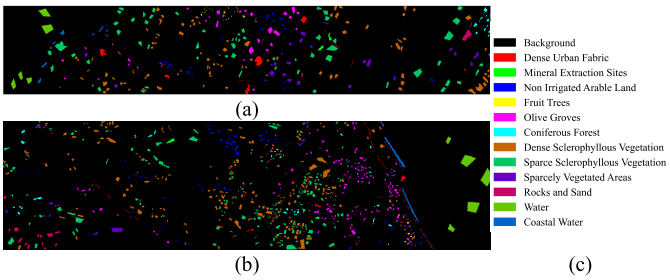



Fig. 7. Ground truths and legend of the HyRANK dataset after processing. (a) Dioni scene. (b) Loukia scene. (c) Legend.

The HyRANK dataset consists of Dioni and Loukia scenes obtained by the EO-1 satellite. Alternatively, the Houston dataset consists of Houston 2013 scene (with a spatial resolution of 2.5 m) and Houston 2018 scene (with a spatial resolution of 1 m) collected by different sensors. The parameters of HyRANK and Houston datasets are listed in Table III and the ground truths are shown in Figs. 7 and 8. There are 12 and seven consistent classes in HyRANK and Houston, respectively, as listed in Table III. Forty-eight bands are extracted from Houston 2013 corresponding to Houston 2018. An overlapping area of 209 × 954 in Houston 2018 is selected. The ground truths and labeled samples of HyRANK and Houston datasets after processing are shown in Figs. 7 and 8, respectively.

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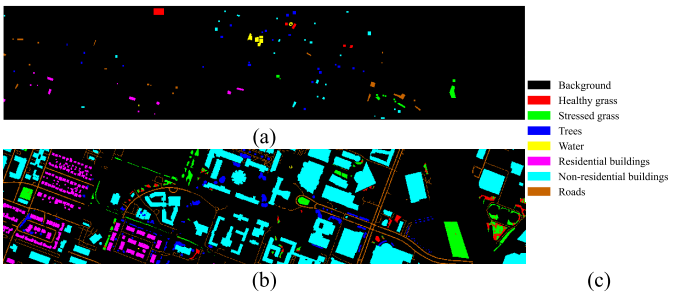



Fig. 8. Ground truths and legend of the Houston dataset after processing. (a) Houston 2013 scene. (b) Houston 2018 scene. (c) Legend.

TABLE III
INFORMATION OF TWO CROSS-SCENE DATASETS

| Name | HyRANK | | Houston | |
|---|---|---|---|---|
| | Dioni | Loukia | Houston 2013 | Houston 2018 |
| Size | 250×1376 | 249×945 | 349×1905 | 601×2384 |
| Bands | 176 | 176 | 144 | 48 |
| Classes | 12 | 12 | 7 | 7 |
| Labeled pixels | 20024 | 13503 | 5070 | 72107 |

### *B. Classification Results*

We designed classification experiments for all selected hyperspectral classification algorithms in Table I using the Indian Pines, Pavia University, and Salinas datasets. Before training, the values in datasets are scaled to 8-bit (0–255) targeting low energy consumption when implemented on-board; 10% of the pixels are randomly selected as training samples in a stratified way. For the MLR, we set the penalty error ($C$) to 10. There are 100 trees in RF and GBDT, as well as RF classifiers in other algorithms. The parameters of ELM, SVM, and CNN2D are the same as in [8]–[10]. Twenty kernels with a size of 24 are used in CNN1D. Thirty-two kernels with a size of $5 \times 5 \times 24$ and 64 kernels with a size of $5 \times 5 \times 16$ are used in two convolutional layers in CNN3D separately. Softmax is used as the activation function in CNN1D, CNN2D, and CNN3D. The structures of HybridSN and $A^2S^2K$-ResNet are also the same as in [27] and [28], respectively. For the LBP-RF, multiscale LBPs are employed on the grayscale image of average values from spectral dimension as same as the dimensionality reduction process of MSLM-RF. For the EMP-RF, the EMP method is the same as in [30]. The whole structure of EMAP-RF is the same as in [31]. The RF is used as the classifier in SPCA-GPs-RF to replace the SVM used in SPCA-GPs [32]. For our MSLM-RF, the parameter of RF in the second stage is the same as the original RF mentioned above. Since the datasets are already part of the actual remote sensing images, we did not slice the three datasets again. The number of spatial feature layers produced by the maximum filter is the smaller value of width and height for each dataset.

After using the trained models to classify the pixels in three datasets, the overall accuracies (OAs), average accuracies (AAs), and Kappa coefficients are listed in Tables IV and V. The classification results are shown in Figs. 9–11. In particular, ELM, SVM, and CNN2D were used in the previous on-board classification works [8]–[10].

The classification accuracies of algorithms only using spectral features in Table IV are significantly lower than the algorithms using spatial and spectral features in Table V. Therefore, joint learning of the spatial and spectral features of hyperspectral images is beneficial to improve the overall classification accuracy of the algorithm. The classification accuracies of MSLM-RF on three datasets are 97.16%, 99.32%, and 99.57%, which are much higher than those by only using spectral features in Table IV.

In Table V, the accuracies of MSLM-RF are only 1.41%, 0.49%, and 0.41% lower than the best results of one-stage spatial–spectral classification algorithms on three datasets. Classification accuracies of the proposed MSLM-RF are close to the ones of the latest state-of-the-art algorithms. The results of two-stage classification algorithms with a similar structure to MSLM-RF are also presented in Table V. MSLM extracts more spatial features by using the pixel value instead of the size relationship of the pixels in the LBP. The classification accuracies of MSLM-RF on three datasets are increased by 7.5%, 1.51%, and 0.34% than LBP-RF. Compared with the EMP and EMAP using PCA, the multiscale method MSLM compensates for the accuracy loss caused by dimensionality reduction via mean calculation and different basic filters. MSLM-RF gets a higher average classification accuracy than EMP-RF. The average accuracy of MSLM-RF is slightly dropped by 0.12% compared to EMAP-RF. Besides, MSLM-RF also maintains similar classification accuracies with a 1.02% loss at most compared with SPCA-GPs-RF using GP due to the selection of the major spatial features of the image by the maximum filters.

We additionally obtained the experimental results using 1% and 5% of the datasets as training samples to verify the impact of training ratio on classification accuracies of different algorithms. As shown in Fig. 12, the classification accuracies of all algorithms evaluated in this work are decreased as the proportion of training samples is reduced from 10% to 1%. For the proposed MSLM-RF, higher classification accuracies are achieved compared to most algorithms with reduced proportion of training samples. When using 1% training samples, only EMAP-RF and SPCA-GPs-RF can simultaneously achieve higher average classification accuracies than MSLM-RF on all the three datasets. However, the energy consumptions of EMAP-RF and SPCA-GPs-RF are significantly higher than that of the proposed MSLM-RF, as discussed in Section III-F. The proposed MSLM-RF can therefore maintain relatively high classification accuracies and low energy consumption with fewer training samples.

For the HyRANK and Houston datasets, we randomly selected 10% training samples in a stratified way. The Loukia scene in the HyRANK dataset and the Houston 2018 scene in the Houston dataset are chosen as the target scenes. Three classification algorithms, including RF, SPCA-GPs-RF, and MSLM-RF, are picked for all using RF as the classifier with low computational complexity. In particular, SPCA-GPs-RF achieves high accuracies with relatively low computational complexities among the spatial–spectral

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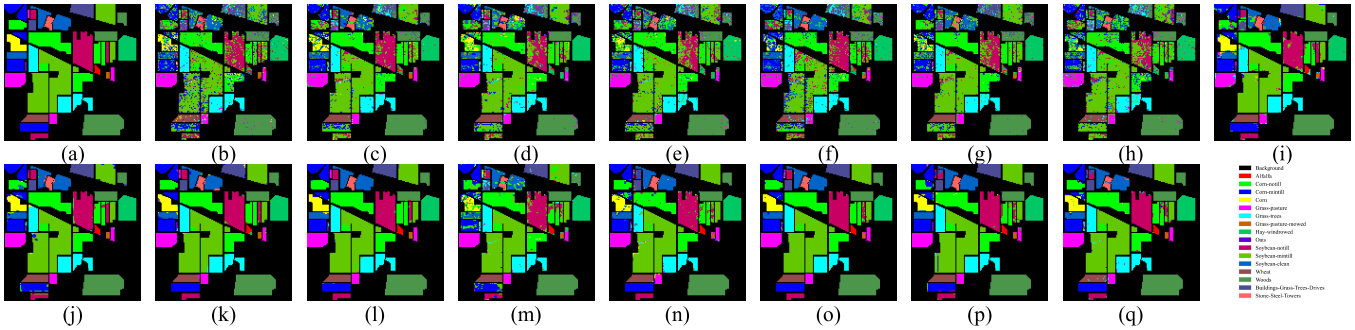



Fig. 9. Classification maps of Indian Pines. (a) Ground Truth. (b) MLR. (c) SVM. (d) RF. (e) GBDT. (f) MLP. (g) CNN1D. (h) ELM. (i) CNN2D. (j) CNN3D. (k) HybridSN [27]. (l) $A^2S^2K$-ResNet [28]. (m) LBP-RF. (n) EMP-RF. (o) EMAP-RF [31]. (p) SPCA-GPs-RF. (q) Proposed MSLM-RF.

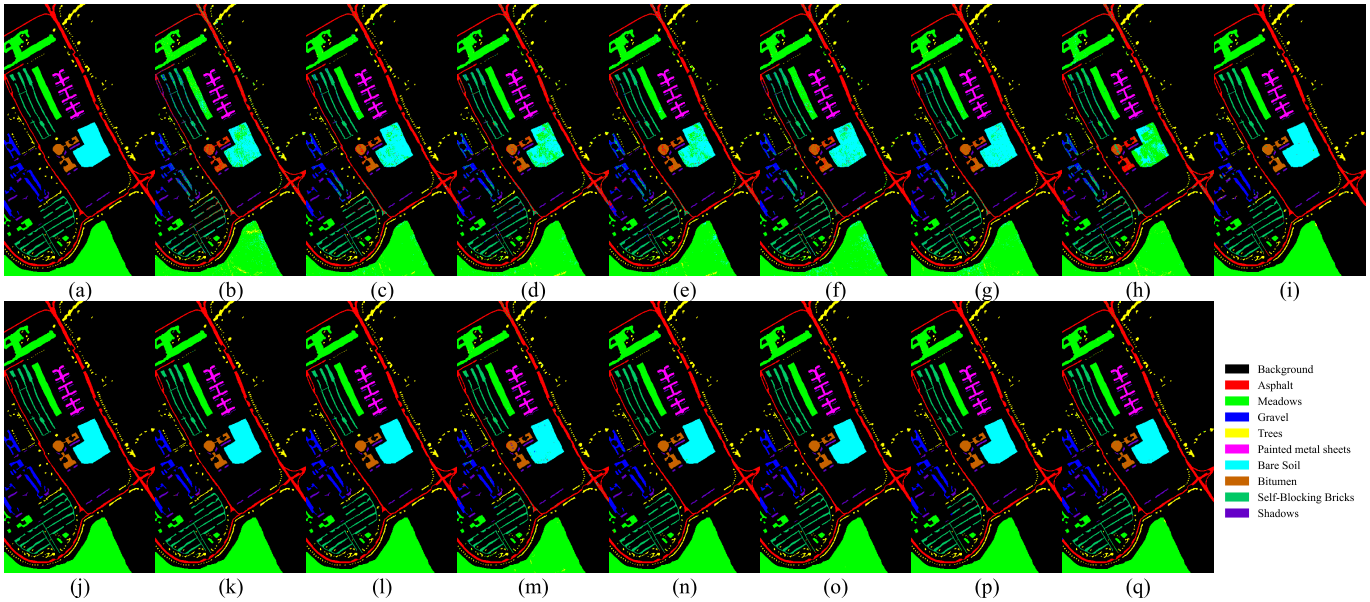



Fig. 10. Classification maps of Pavia University. (a) Ground Truth. (b) MLR. (c) SVM. (d) RF. (e) GBDT. (f) MLP. (g) CNN1D. (h) ELM. (i) CNN2D. (j) CNN3D. (k) HybridSN [27]. (l) $A^2S^2K$-ResNet [28]. (m) LBP-RF. (n) EMP-RF. (o) EMAP-RF [31]. (p) SPCA-GPs-RF. (q) Proposed MSLM-RF.

TABLE IV
CLASSIFICATION ACCURACIES OF SPECTRAL ALGORITHMS USING 10% SAMPLES PER CLASS ON THREE DATASETS (100×)

| | | MLR | SVM [9] | RF | GBDT | MLP | CNN1D | ELM [8] | **MSLM-RF** |
|---|---|---|---|---|---|---|---|---|---|
| Indian Pines | OA | 76.23 | 82.34 | 77.62 | 73.73 | 73.80 | 80.11 | 67.87 | 97.16 |
| | AA | 72.37 | 84.39 | 82.53 | 63.45 | 73.96 | 86.72 | 72.12 | 96.26 |
| | Kappa | 72.88 | 79.75 | 74.26 | 69.67 | 70.28 | 77.01 | 62.77 | 96.76 |
| Pavia University | OA | 86.34 | 94.41 | 90.56 | 91.03 | 90.95 | 95.07 | 82.66 | 99.32 |
| | AA | 83.27 | 94.09 | 91.59 | 90.78 | 91.20 | 94.06 | 85.33 | 99.44 |
| | Kappa | 81.82 | 92.56 | 87.26 | 87.96 | 87.95 | 93.48 | 75.97 | 99.10 |
| Salinas | OA | 91.53 | 93.55 | 92.27 | 91.60 | 90.11 | 94.62 | 89.37 | 99.57 |
| | AA | 96.16 | 96.84 | 95.43 | 95.47 | 94.33 | 97.72 | 93.57 | 99.44 |
| | Kappa | 90.55 | 92.81 | 91.38 | 90.63 | 88.96 | 94.00 | 88.13 | 99.52 |

classification algorithms compared above. The classification results of different combinations of data processing (L1 and ST) and domain adaptation (CORAL) methods on the targeted scenes of HyRANK and Houston datasets are listed in Tables VI and VII, respectively. The number of spatial feature layers generated by MSLM is optimized for achieving high classification results of the proposed MSLM-RF algorithm, as labeled in parentheses in Tables VI and VII.

### C. Cross-Scene Classification Performance of MSLM-RF

In the real-world hyperspectral on-board classification application, the model trained on samples from the source scenes

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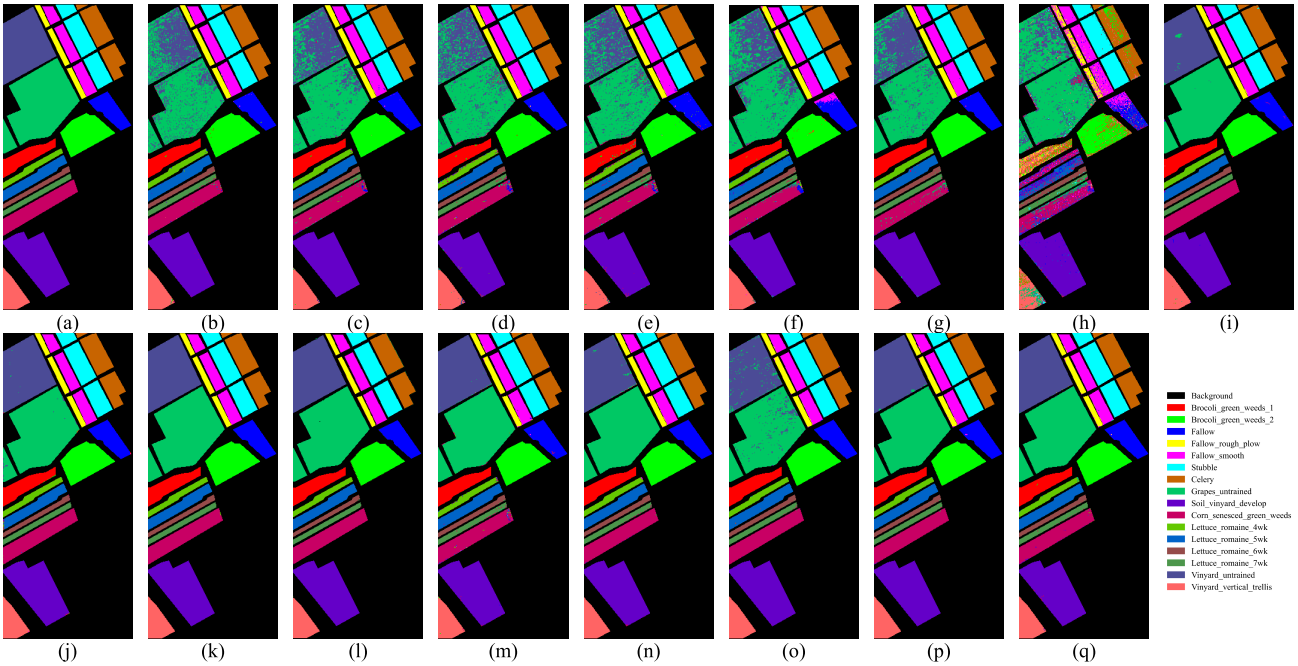



Fig. 11. Classification maps of Salinas. (a) Ground Truth. (b) MLR. (c) SVM. (d) RF. (e) GBDT. (f) MLP. (g) CNN1D. (h) ELM. (i) CNN2D. (j) CNN3D. (k) HybridSN [27]. (l) $A^2S^2K$-ResNet [28]. (m) LBP-RF. (n) EMP-RF. (o) EMAP-RF [31]. (p) SPCA-GPs-RF. (q) Proposed MSLM-RF.

TABLE V

CLASSIFICATION ACCURACIES OF SPATIAL–SPECTRAL ALGORITHMS USING 10% SAMPLES PER CLASS ON THREE DATASETS (100×)

| | | One-stage algorithms | | | | Two-stage algorithms | | | | **MSLM-RF** |
|---|---|---|---|---|---|---|---|---|---|---|
| | | CNN2D [10] | CNN3D | HybridSN [27] | $A^2S^2K$-ResNet [28] | LBP-RF [29] | EMP-RF [30] | EMAP-RF [31] | SPCA-GPs-RF [32] | |
| Indian Pines | OA | 93.43 | 96.55 | 97.44 | 98.57 | 89.66 | 95.15 | 98.71 | 98.18 | 97.16 |
| | AA | 93.24 | 96.66 | 93.47 | 97.20 | 91.59 | 95.76 | 99.18 | 98.38 | 96.26 |
| | Kappa | 92.50 | 96.06 | 97.08 | 98.37 | 88.14 | 94.46 | 98.53 | 97.92 | 96.76 |
| Pavia University | OA | 98.58 | 99.89 | 99.72 | 99.81 | 97.81 | 99.71 | 99.64 | 99.83 | 99.32 |
| | AA | 97.87 | 99.82 | 99.11 | 99.77 | 97.79 | 99.71 | 99.76 | 99.81 | 99.44 |
| | Kappa | 98.12 | 99.85 | 99.63 | 99.75 | 97.10 | 99.62 | 99.53 | 99.77 | 99.10 |
| Salinas | OA | 99.62 | 99.62 | 99.98 | 99.91 | 99.23 | 99.57 | 98.06 | 99.84 | 99.57 |
| | AA | 99.53 | 99.54 | 99.98 | 99.90 | 98.87 | 99.60 | 98.94 | 99.82 | 99.44 |
| | Kappa | 99.57 | 99.58 | 99.98 | 99.90 | 99.14 | 99.52 | 97.84 | 99.82 | 99.52 |

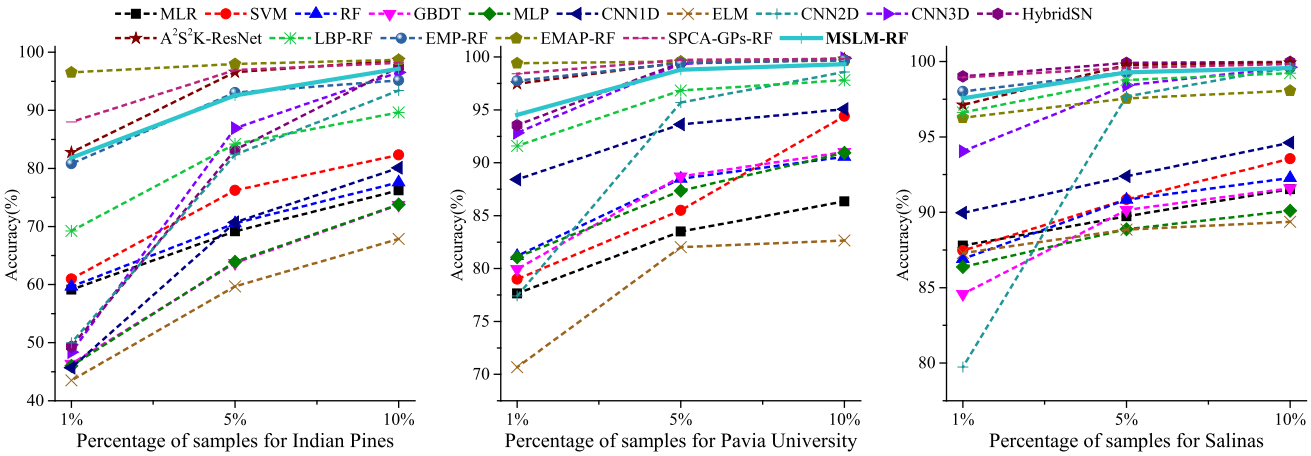



Fig. 12. Classification accuracies using 1%, 5%, and 10% samples on three datasets.

will be used to classify the newly acquired target scenes after loaded onto the satellite. Spectral drift and distortion would occur between the source and target scenes due to the atmospheric and light conditions, sensor nonlinearities, different material compositions of the same land cover category, and so on [42].

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TABLE VI
CLASSIFICATION ACCURACIES USING 10% SAMPLES PER CLASS ON THE TARGET SCENE LOUKIA OF THE HYRANK DATASET (100×)

| | RF | | | | | SPCA-GPs-RF | | | | | **MSLM-RF** | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | w/o | L1 | ST | L1+CORAL | **ST+CORAL** | w/o | **L1** | ST | L1+CORAL | ST+CORAL | w/o (8) | L1 (96) | ST (114) | L1+CORAL (96) | **ST+CORAL (76)** |
| OA | 46.66 | 51.99 | 55.45 | 52.08 | **58.59** | 15.51 | **43.38** | 31.27 | 42.60 | 36.62 | 56.74 | 55.22 | 57.65 | 55.46 | **59.95** |
| AA | 27.42 | 50.12 | 42.85 | 51.38 | **54.10** | 37.53 | **25.98** | 28.59 | 26.26 | 26.24 | 44.62 | 50.50 | 48.42 | 53.00 | **52.22** |
| Kappa | 33.13 | 41.27 | 45.95 | 41.41 | **49.92** | -3.67 | **26.95** | 16.97 | 25.80 | 23.65 | 47.19 | 44.44 | 48.29 | 44.77 | **51.06** |

w/o: without applying any data processing or domain adaptation methods for original classification algorithms.

TABLE VII
CLASSIFICATION ACCURACIES USING 10% SAMPLES PER CLASS ON THE TARGET SCENE HOUSTON 2018 OF THE HOUSTON DATASET (100×)

| | RF | | | | | SPCA-GPs-RF | | | | | **MSLM-RF** | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | w/o | L1 | ST | **L1+CORAL** | ST+CORAL | w/o | **L1** | ST | L1+CORAL | ST+CORAL | w/o (101) | L1 (34) | ST (143) | **L1+CORAL (34)** | ST+CORAL (114) |
| OA | 9.30 | 65.03 | 30.13 | **65.03** | 24.73 | 34.41 | **39.36** | 34.63 | 39.35 | 34.73 | 31.15 | 68.56 | 31.65 | **68.76** | 29.55 |
| AA | 1.33 | 61.44 | 17.86 | **61.44** | 15.59 | 37.76 | **55.95** | 21.89 | 55.89 | 32.66 | 18.73 | 63.96 | 18.08 | **64.03** | 15.90 |
| Kappa | 0.00 | 47.91 | 11.01 | **47.91** | 6.85 | 18.45 | **23.11** | 12.18 | 23.10 | 18.30 | 12.51 | 51.36 | 12.17 | **51.54** | 8.08 |

w/o: without applying any data processing or domain adaptation methods for original classification algorithms.

To eliminate the spectral variations between different scenes, data processing and domain adaptation methods [43] are typically required in company with the hyperspectral on-board classification algorithms to obtain better cross-scene classification results. Normalization and standardization are two commonly used data processing methods in image classification. When used for hyperspectral images, the L1 normalization method (L1) can mitigate the amplitude shifts in the spectral vectors between different scenes [44]. The standardization method (ST) also mitigates the amplitude variations and further reduces the impact of spectral vector variances in different scenes on algorithm training. The L1 normalization and standardization methods are computationally efficient, which results in small energy consumption increment for on-board cross-scene classification. On the other hand, correlation alignment (CORAL) [43] is selected as the domain adaptation method for its low computational complexity, as well as low energy consumption for on-board classification. CORAL reduces the domain shift by aligning the covariances of source and target domain distributions [43]. CORAL is stable for hyperspectral classification for its zero hyperparameter.

For the results of HyRANK dataset as listed in Table VI, the classification accuracy of RF is effectively increased by 8.79% with ST method (ST+RF) while only improving by 5.33% with L1 normalization (L1+RF). The classification accuracy of ST+RF is further improved by 3.14% after applying the CORAL method (ST+CORAL+RF). The accuracies can also be improved for SPCA-GPs-RF and the proposed MSLM-RF after applying the L1/ST and CORAL methods in Table VI. Specifically, ST+CORAL+MSLM-RF with 76 layers of spatial features gets the highest classification accuracy of 59.95%, which is 1.36% and 16.57% higher than the accuracies of ST+CORAL+RF and L1+SPCA-GPs-RF, respectively.

For the results of Houston dataset as listed in Table VII, the classification accuracy of RF with L1 normalization (L1+RF) is 34.9% higher than that of RF with standardization (ST+RF).

The classification accuracy of MSLM-RF is also improved by 37.41% after employing L1 normalization and further increased by 0.2% when using the CORAL method. In particular, L1+CORAL+MSLM-RF with 34 layers of spatial features gets the highest classification accuracy of 68.76%, which is 3.73% and 29.4% higher than the accuracies of L1+CORAL+RF and L1+SPCA-GPs-RF, respectively.

From the above experimental results, the MSLM-RF algorithm can still achieve high accuracies in the cross-scene classification on both HyRANK and Houston datasets after applying the data processing and domain adaptation methods. Furthermore, even applying the same data processing or domain adaptation methods, the accuracy of the proposed MSLM-RF is also higher than other algorithms from Tables IV and V, indicating the effectiveness of the proposed MSLM-RF.

### *D. Effectiveness Verification of MSLM*

The impacts of different hyperparameters and settings of MSLM still need to be verified through experiments on the Indian Pines, Pavia University, and Salinas datasets. Besides, we compared the classification results of different combinations of feature extraction methods and classifiers to further measure the effectiveness of the spatial feature extraction method proposed.

As mentioned in Section II, the coordinates of the center points are calculated approximately for maximum filters with even window sizes in MSLM. The positions of filters with different sizes are not aligned, as shown in Fig. 4. By selecting pixels in the corner of the filter window, misalignment can be avoided and different classification accuracies will be obtained.

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TABLE VIII
CLASSIFICATION ACCURACIES USING DIFFERENT REFERENCE POINTS ON THREE DATASETS (100×)

| | | Center | Top Left | Top Right | Bottom Left | Bottom Right |
|---|---|---|---|---|---|---|
| Indian Pines | OA | **97.16** | 96.22 | 94.10 | 94.37 | 93.24 |
| | AA | **96.26** | 95.18 | 94.22 | 94.29 | 94.07 |
| | Kappa | **96.76** | 95.69 | 93.26 | 93.57 | 92.28 |
| Pavia University | OA | 99.32 | 99.10 | **99.38** | 99.12 | 99.14 |
| | AA | 99.44 | 99.19 | **99.46** | 99.14 | 99.21 |
| | Kappa | 99.10 | 98.81 | **99.18** | 98.83 | 98.86 |
| Salinas | OA | **99.57** | 99.38 | 99.55 | 99.38 | 99.40 |
| | AA | **99.44** | 99.22 | 99.40 | 99.15 | 99.26 |
| | Kappa | **99.52** | 99.31 | 99.50 | 99.31 | 99.33 |

TABLE IX
COMPARISON OF MSLM-RFS WITH MAXIMUM AND MINIMUM FILTERS UNDER DIFFERENT NOISE CONDITIONS (100×)

| | | Maximum | Minimum |
|---|---|---|---|
| Indian Pines | No noise | 97.16 | 93.98 |
| | 36.38dB | 95.72 | 94.02 |
| | 26.38dB | 95.87 | 93.85 |
| Pavia University | No noise | 99.32 | 99.11 |
| | 35.07dB | 99.30 | 98.89 |
| | 25.07dB | 99.36 | 98.09 |
| Salinas | No noise | 99.57 | 99.26 |
| | 35.07dB | 99.51 | 99.07 |
| | 25.07dB | 99.56 | 99.44 |

Therefore, we get the classification results on three datasets by using the center, top left, top right, bottom left, and bottom right pixels in MSLM separately. The accuracies are listed in Table VIII and varied slightly according to different reference pixels selected. The highest average classification accuracy is obtained by using the center pixel. The accuracies are affected by the texture of the hyperspectral image since the order of the classification accuracies is independent of the selected point.

As discussed in Section II, using maximum values rather than minimum values to represent texture features in MSLM is less susceptible to noise. To verify the antinoise ability of the maximum filter used as the basic operating unit in MSLM, the Gaussian noise is artificially added to the hyperspectral datasets [45]. The basic operation unit in MSLM-RF is also changed to the minimum filter to compare with the original MSLM-RF. The OA values of classification results in different situations are listed in Table IX. MSLM-RF using the maximum filters obtains higher classification accuracies when no noise is added to the datasets. The extracted maximum values better represent the spatial features of the datasets. As the noise level increases, the classification accuracies obtained by using the maximum filters are still significantly higher than the accuracies using the minimum filters. In the Pavia University and Salinas datasets, a small amount of noise does not cause a large reduction in the classification accuracies of MSLM-RF with maximum filters.

Two extra comparative experiments are designed for different combinations of feature extraction methods and classifiers. The typical classifiers used by the algorithms in Tables IV and V are SVM, RF, and the last fully connected layer of CNNs. In the first experiment, the RF classifier in MSLM-RF is replaced by SVM (MSLM-SVM) and the fully connected layer (MSLM-FC). The classification results of MSLM-SVM and MSLM-FC are shown in Table X. When using SVM as the classifier, MSLM-SVM can still effectively extract spatial features and obtain classification accuracies close to those of MSLM-RF. The accuracies are much higher than the results of only using SVM for classification in Table IV. The classification accuracies of MSLM-FC using the fully connected layer classifier are much lower than

TABLE X
COMPARISON OF MSLM AND DIFFERENT CLASSIFIERS COMBINATIONS (100×)

| | | MSLM-SVM | MSLM-FC | **MSLM-RF** |
|---|---|---|---|---|
| Indian Pines | OA | 95.38 | 82.80 | **97.16** |
| | AA | 96.15 | 81.95 | **96.26** |
| | Kappa | 94.72 | 80.35 | **96.76** |
| Pavia University | OA | **99.47** | 89.94 | 99.32 |
| | AA | **99.50** | 87.54 | 99.44 |
| | Kappa | **99.30** | 86.80 | 99.10 |
| Salinas | OA | 99.53 | 94.21 | **99.57** |
| | AA | 99.56 | 96.35 | **99.44** |
| | Kappa | 99.48 | 93.56 | **99.52** |

MSLM-RF. A single fully connected layer has inferior classification ability than SVM and RF.

In the second experiment, the classification accuracies of different CNN feature extraction methods combined with RF are tested. CNN1D and CNN2D are selected from the spectral classification algorithms and the one-stage spatial–spectral classification algorithms, respectively. The last fully connected layers in CNN1D and CNN2D are replaced with the RF classifier (CNN1D-RF and CNN2D-RF). The classification results are shown in Table XI. When the same RF classifier is used, the accuracies of MSLM-RF are higher than CNN1D-RF and CNN2D-RF. The feature extraction ability of MSLM is better than that of CNN convolutional layers with a simple structure. By comprehensively comparing the classification results in Tables V, X, and XI, the above experiments have proved the effectiveness of the proposed spatial feature extraction method.

### *E. Number of Operations in the Inference*

Only a few previous works considered the hardware implementation of the algorithm. An overview of inference properties of several algorithms is provided in [46] and the model sizes and numbers of operations are evaluated. By comparing the proposed MSLM-RF with other algorithms, we also get the differences in the types and numbers of operations, as shown

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TABLE XI
COMPARISON OF DIFFERENT CNN FEATURE EXTRACTION METHODS AND RF COMBINATIONS (100×)

| | | CNN1D-RF | CNN2D-RF | **MSLM-RF** |
|---|---|---|---|---|
| Indian Pines | OA | 80.88 | 92.53 | **97.16** |
| | AA | 87.78 | 93.65 | **96.26** |
| | Kappa | 77.93 | 91.44 | **96.76** |
| Pavia University | OA | 95.37 | 97.93 | **99.32** |
| | AA | 95.19 | 97.79 | **99.44** |
| | Kappa | 93.83 | 97.25 | **99.10** |
| Salinas | OA | 95.15 | 99.19 | **99.57** |
| | AA | 97.67 | 99.45 | **99.44** |
| | Kappa | 94.59 | 99.10 | **99.52** |

in Fig. 13. The types of operation include integer comparison (INT Cmp), floating-point addition (FP Add), floating-point multiplication (FP Mul), and floating-point exponentiation (FP Exp). The numbers of operations come from the formulas of different algorithms in Table I, based on the actual inference process on the Indian Pines, Pavia University, and Salinas datasets. The software optimizations of the algorithms are not considered. The values in Fig. 13 are the average numbers of operations performed on each pixel.

As can be seen from Fig. 13(a), SVM requires the largest number of operations in spectral classification algorithms and contains exponential operations. MLR and RF models require fewer operations during inference than other spectral algorithms. MLP, CNN1D, and ELM need more operations than any spectral classification algorithms except for SVM and they all consist of floating-point multiplication and accumulation.

In Fig. 13(b), the numbers of operations of the MSLM-RF algorithm are about $10^5$, $10^4$, and $10^6$ times less than those of CNN3D, HybridSN, and A$^2$S$^2$K-ResNet, respectively. Compared with the on-board classification algorithm CNN2D, MSLM-RF reduces the number of operations by about $10^3$ times and gets higher average classification accuracy.

In the two-stage algorithms with joint learning of spatial–spectral features, LBP-RF uses only a fixed number of pixels in each LBP operator to compare with the center pixel. The number of operations in LBP-RF is about twice as much as RF and lower than MSLM-RF. Either PCA or LDA takes a large portion of operations in EMP-RF and EMAP-RF. Consequently, both EMP-RF and EMAP-RF need substantially larger numbers of operations than MSLM-RF. The operations of SPCA-GPs-RF are also mainly occupied by the SPCA and consist of multiplications and additions. MSLM-RF still gets a smaller number of operations in Indian Pines and Salinas datasets.

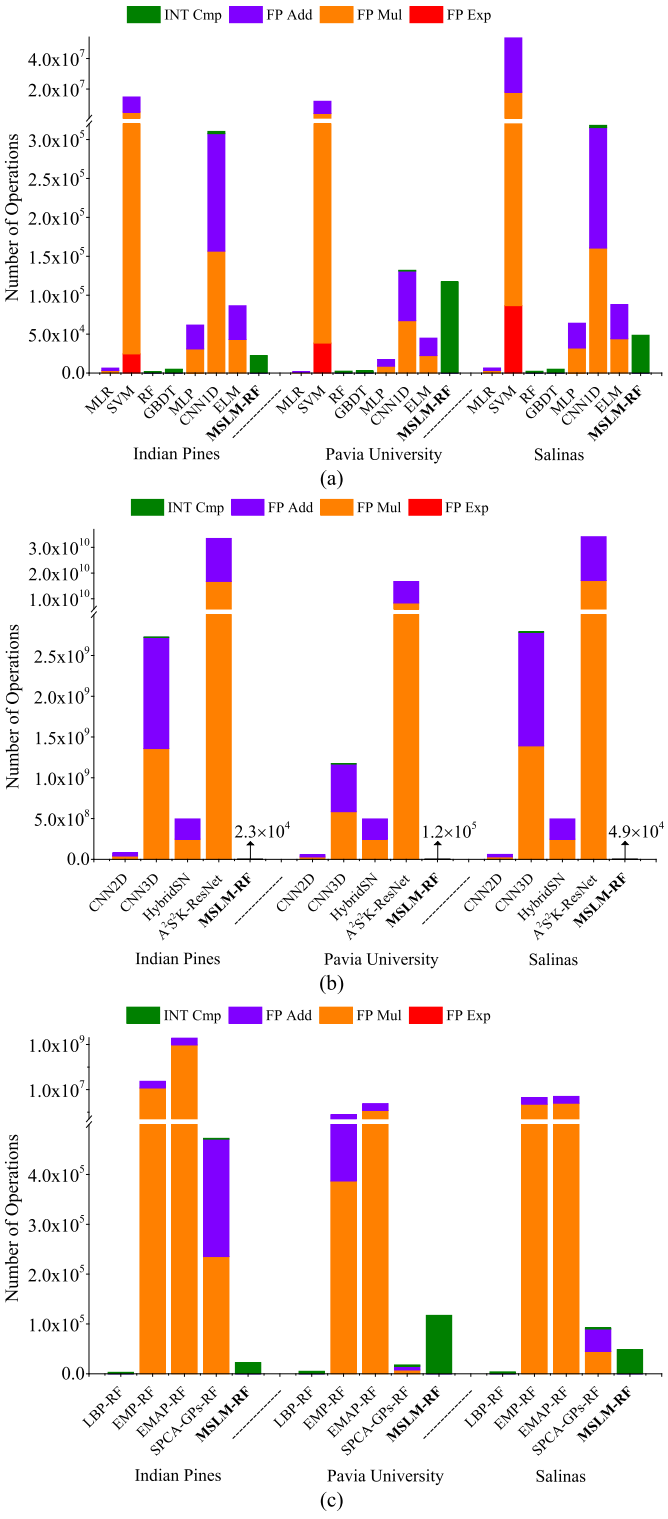



Fig. 13. Numbers of operations in the inference stage of different algorithms. (a) Spectral algorithms. (b) One-stage spatial–spectral algorithms. (c) Two-stage spatial–spectral algorithms. Different colors in the stacked histograms represent different calculation types (integer comparison, floating-point addition, floating-point multiplication, and floating-point exponentiation).

### *F. Energy Consumptions in the Inference*

In the inference process of different algorithms, the data precisions of floating-point multiplication and addition are 32 bit. The precision of floating-point exponentiation in SVM is also 32 bit and the base of exponential operation is the fixed value $e$. The integer comparison operations are mainly used in RF, GBDT, and algorithms using RF as the classifier (including MSLM-RF). The data precision of integer comparison is 8 bit, which is the same as the input data. The energy consumption of floating-point addition and multiplication is provided in [47] and the data are listed in Table XII. The

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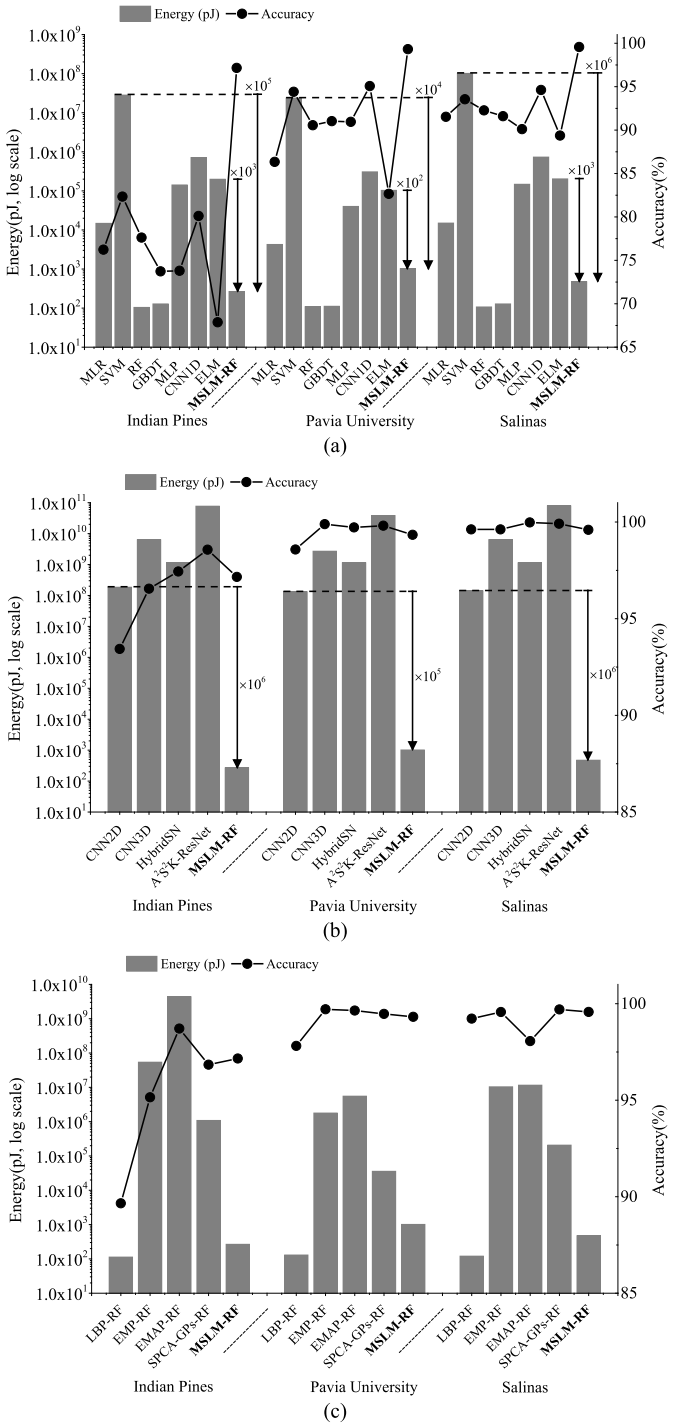



Fig. 14. Energy in the inference stage and corresponding accuracies (10% training samples) of different algorithms. (a) Spectral algorithms. (b) One-stage spatial–spectral algorithms. (c) Two-stage spatial–spectral algorithms. The lines represent the classification accuracies of algorithms on three datasets. The differences in energy consumption between MSLM-RF and previous on-board algorithms are shown in (a) and (b).

energy consumption of integer comparison and floating-point exponentiation is obtained from circuit simulations in a similar environment to the 40-nm chip process. Finally, we get the energy consumption of each algorithm in the inference process, which is listed in Table XIII and shown in Fig. 14.

TABLE XII
ENERGY CONSUMPTIONS OF VARIOUS OPERATIONS [43]

| Type | Precision | Energy |
|---|---|---|
| Integer Comparison | 8-bit | 0.008pJ |
| Floating-point Addition | 32-bit | 0.9pJ |
| Floating-point Multiplication | 32-bit | 3.7pJ |
| Floating-point Exponentiation | 32-bit | 38.975pJ |

TABLE XIII
ENERGY CONSUMPTIONS OF EACH ALGORITHM IN THE INFERENCE STAGE (PJ)

| Algorithms | Indian Pines | Pavia University | Salinas |
|---|---|---|---|
| MLR | 1.5E+04 | 4.3E+03 | 1.5E+04 |
| SVM | 2.8E+07 | 2.4E+07 | 1.0E+08 |
| RF | 1.0E+02 | 1.1E+02 | 1.1E+02 |
| GBDT | 1.3E+02 | 1.1E+02 | 1.3E+02 |
| MLP | 1.4E+05 | 4.0E+04 | 1.5E+05 |
| CNN1D | 7.2E+05 | 3.1E+05 | 7.3E+05 |
| ELM | 2.0E+05 | 1.0E+05 | 2.0E+05 |
| CNN2D | 1.8E+08 | 1.3E+08 | 1.4E+08 |
| CNN3D | 6.3E+09 | 2.7E+09 | 6.4E+09 |
| HybridSN | 1.1E+09 | 1.1E+09 | 1.1E+09 |
| $A^2S^2K$-ResNet | 7.7E+10 | 3.8E+10 | 7.9E+10 |
| LBP-RF | 1.1E+02 | 1.3E+02 | 1.2E+02 |
| EMP-RF | 5.4E+07 | 1.8E+06 | 1.0E+07 |
| EMAP-RF | 4.4E+09 | 5.5E+06 | 1.2E+07 |
| SPCA-GPs-RF | 1.1E+06 | 3.6E+04 | 2.1E+05 |
| **MSLM-RF** | 2.7E+02 | 1.0E+03 | 4.7E+02 |

In Table XIII and Fig. 14, only RF, GBDT, and LBP-RF have lower energy consumptions than MSLM-RF, but their classification accuracies are reduced by more than 7.5%. The one-stage spatial–spectral algorithms have very high energy consumptions and are therefore unsuitable for on-board classification. The classification accuracies of MSLM-RF are only reduced by 1.41% at most than those of CNN3D, HybridSN, and $A^2S^2K$-ResNet. The energy consumptions of the two-stage algorithms except LBP-RF are more than 30 times higher than those of MSLM-RF. From Table XIII and Fig. 14, MSLM-RF gets a relatively low energy consumption among previous algorithms of different categories.

Compared to the existing on-board classification works [8]–[10], the energy consumed by the proposed MSLM-RF is more than 100 times lower than the algorithms used in these studies (ELM [8], SVM [9], and CNN2D [10]). MSLM-RF is, therefore, suitable for on-board hyperspectral classification tasks. The FPGA and GPU hardware environments used in [8]–[10] can also be used to execute MSLM-RF on-board.

The spatial feature extraction method we proposed is based on maximum filters and requires just integer comparison operations. Moreover, the energy consumption of MSLM-RF in the classification process will be lower than previous

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TABLE XIV
CLASSIFICATION ACCURACY OF MSLM-RF-NOL USING 10% SAMPLES PER CLASS ON THREE DATASETS (100×)

| | Indian Pines | Pavia University | Salinas |
|---|---|---|---|
| OA | 87.56 | 94.80 | 98.63 |
| AA | 89.93 | 94.72 | 98.55 |
| Kappa | 85.75 | 93.07 | 98.48 |

experimental results if the calculation process is further optimized at the software level. The maximum filtering result of each pixel can be obtained through six comparisons regardless of the window size [48]. The generation of multiscale spatial feature layers can also be completed in parallel. For on-board classification tasks, MSLM-RF can greatly reduce energy consumption when classifying a large amount of hyperspectral data. On hyperspectral satellites with a strict energy budget, MSLM-RF can accomplish more scheduled monitoring and warning tasks than previous on-board algorithms.

### *G. Extension Algorithms of MSLM-RF With Evaluation in Accuracy and Computational Complexity*

When the power consumption requirements of the on-board implementation are extremely low, the proposed MSLM-RF can be adjusted to further reduce the number of operations and corresponding energy consumption at the cost of reducing the classification accuracy. In addition to the common pruning and low-precision quantification methods, there are also some optimization methods specifically applicable to MSLM-RF. These methods can be employed according to the actual accuracy and energy consumption requirements of specific on-board classification tasks. The energy consumptions of the extension algorithms are not calculated since the operation types are the same as MSLM-RF.

Window sliding in the spatial feature extraction stage of MSLM-RF causes lots of data to be repetitively compared. As shown in Fig. 15(a), using nonoverlap maximum filter windows rather than sliding windows can significantly reduce the repeated calculations of data. All pixels are replaced with the same maximum value in each window to generate spatial feature layers. Due to fixed window positions of nonoverlap filters, the maximum results in larger filter windows can be calculated by reusing the result of smaller filters in the same position.

The OAs, AAs, and Kappa coefficients of the algorithm using nonoverlap maximum filters (MSLM-RF-NOL) on the Indian Pines, Pavia University, and Salinas datasets are listed in Table XIV. The numbers of operations in the inference stage are shown in Fig. 16. The numbers of operations of MSLM-RF-NOL are virtually the same as those of a single RF, but the classification accuracies on three datasets are 4%–10% higher. Besides, the numbers of operations of MSLM-RF-NOL are much smaller than those of MSLM-RF.

Using the maximum filters with a larger size increment is another way to reduce the computational complexity of the spatial feature extraction method. As shown in Fig. 15(b), the

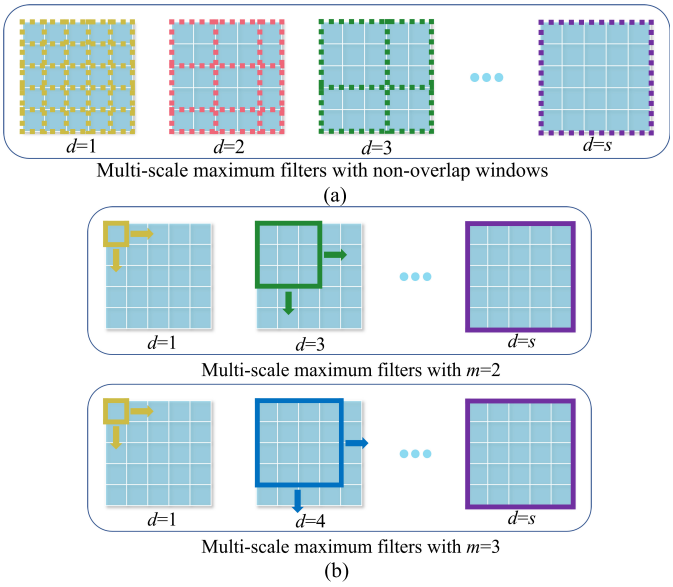



Fig. 15. (a) Extension algorithm of MSLM-RF using nonoverlap maximum filter windows. (b) Extension algorithm of MSLM-RF using different intervals of filter window sizes.

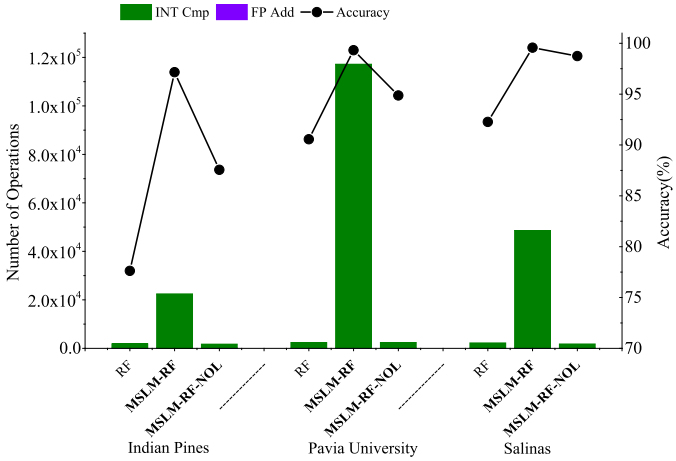



Fig. 16. Numbers of operations in the inference stage and corresponding accuracies (10% training samples) of RF, MSLM-RF, and MSLM-RF-NOL.

increment of window sizes $m$ between the adjacent filters can be adjusted to 2 or more instead of 1 used in the MSLM-RF. The number of spatial feature layers and the length of spatial feature vectors decrease to $1/m$ compared to MSLM-RF as the interval of filter window sizes increases.

The classification accuracies of the algorithm using larger intervals of window sizes (MSLM-RF-SEL) on three datasets are listed in Table XV. The number of operations required in the inference stage of MSLM-RF-SEL remains unchanged since all values in the largest filter window still need to be compared for an individual pixel. Provided that software-level methods [48] are used to optimize the calculation process, the number of operations will be further reduced to approximately $1/m$ of that in the MSLM-RF.

In summary, MSLM-RF-NOL achieves the lowest computational complexity and maintains high classification accuracies, while MSLM-RF-SEL makes a slight reduction in classification accuracy to significantly lower the number of

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TABLE XV
CLASSIFICATION ACCURACY OF MSLM-RF-SEL WITH DIFFERENT INTERVALS OF WINDOW SIZES USING 10% SAMPLES PER CLASS ON THREE DATASETS (100×)

| | | $m$=2 | $m$=3 | $m$=4 | $m$=5 |
|---|---|---|---|---|---|
| Indian Pines | OA | 96.26 | 95.30 | 94.52 | 93.96 |
| | AA | 95.96 | 95.09 | 94.34 | 93.64 |
| | Kappa | 95.73 | 94.63 | 93.73 | 93.10 |
| Pavia University | OA | 99.31 | 99.30 | 99.28 | 99.20 |
| | AA | 99.37 | 99.39 | 99.36 | 99.31 |
| | Kappa | 99.08 | 99.07 | 99.04 | 98.94 |
| Salinas | OA | 99.56 | 99.48 | 99.46 | 99.41 |
| | AA | 99.38 | 99.22 | 99.20 | 99.16 |
| | Kappa | 99.51 | 99.42 | 99.40 | 99.35 |

operations and energy consumption compared to MSLM-RF. MSLM-RF-NOL and MSLM-RF-SEL can also provide compromise solutions to on-board classification tasks. The MSLM-RF-SEL is better than MSLM-RF when a little percent reduction in classification accuracy is accepted. The MSLM-RF-NOL can be used in extremely low power conditions to replace the algorithms such as RF and gets better classification results.

## IV. CONCLUSION

In this article, we propose a new hyperspectral land cover classification algorithm named MSLM-RF. The framework consists of multiscale maximum filters for spatial feature extraction and RF for classification. MSLM-RF gives primary attention to the hardware cost and energy consumption in the inference stage for on-board classification. As a result, MSLM-RF achieves the classification accuracies of 97.16%, 99.32%, and 99.57% on the Indian pines, Pavia University, and Salinas datasets, respectively. The accuracy loss of MSLM-RF is kept within 1.5% when compared to state-of-the-art convolutional neural networks. The energy consumption of MSLM-RF in classification is more than 30 times lower than spatial–spectral algorithms with similar accuracies. Besides, MSLM-RF improves the average classification accuracy higher than 1.47% on three datasets and reduces the energy consumption by more than 100 times compared with previous on-board classification algorithms. MSLM-RF also gets relatively high cross-scene classification accuracies on the HyRANK and Houston datasets. Benefiting from the small number of 8-bit integer comparisons and additions, MSLM-RF greatly reduces the energy consumption in on-board classification and provides a good solution for real-time remote sensing tasks such as monitoring natural disasters or water pollution. Meanwhile, MSLM-RF not only reduces the time to accomplish classification and corresponding tasks but also effectively alleviates the tight transmission bandwidth and storage capacity of hyperspectral data under strict energy restraint.

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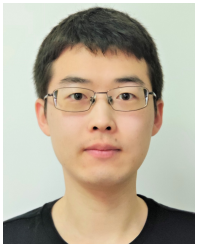
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